

Probability & Statistics

BCA — Semester III — 2021 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

11. Discuss the role of statistics in computer application.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. The following table shows the monthly salary of employees in a certain locality of Lalitpur Sub-metropolitan City:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. Define the term correlation. Compute Karl Pearson's coefficient of correlation between advertisement cost and sales as per the data given below:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. The following table gives the age of the computers of a certain company and annual maintenance costs:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. Fit a Poisson distribution of the following data and calculate the expected frequencies:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. Determine First Quartile (Q_1), D_7 and P_{80} (80th percentile) from the following data:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

17. A box contains 30 items of which 10 are defectives. If two items are selected randomly from the box without replacement, what is the probability that (a) Both are defective (b) Both are non-defective

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

18. The following data represents the scores made in an intelligence test by the groups of students from section A and section B of National College:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. The lifetime of a certain electronic component is a normal random variate with an expectation of 5000 hours and a standard deviation of 100 hours. Compute the probabilities under the following conditions:
a) Lifetime of components is less than 3012 hours
b) Lifetime of components between 4000 to 6000 hours
c) Lifetime of components less than 4500 hours
d) Lifetime of components more than 7000 hours

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. Perform the ANOVA from Two-Way classification:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.